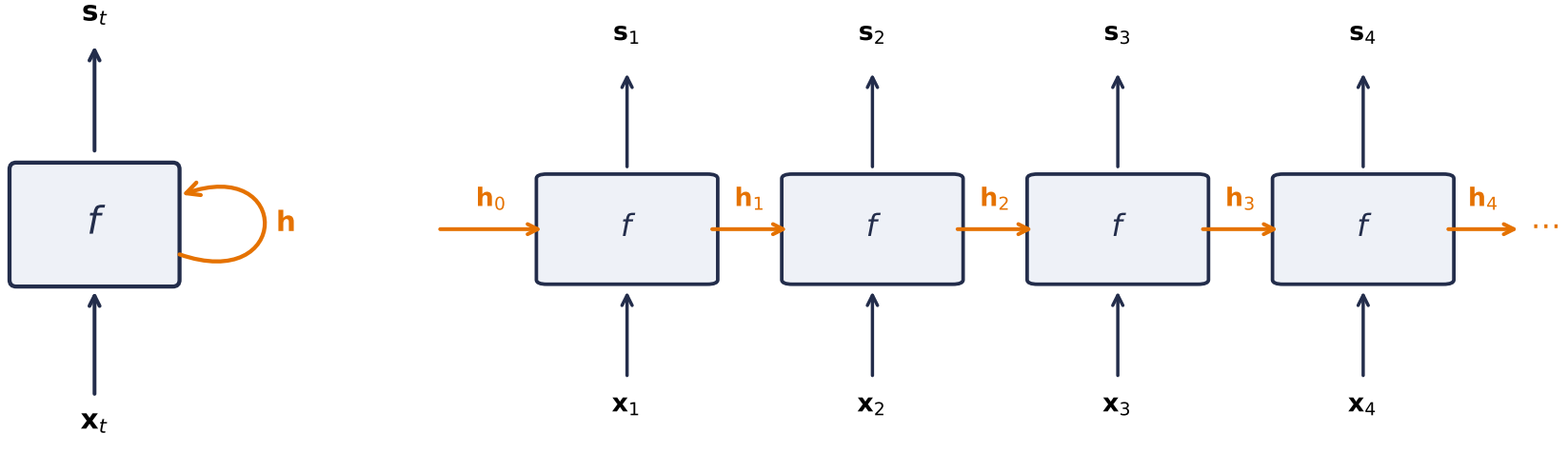
one cell, with a loop
$\mathbf{s}_t$
f
$\mathbf{h}$
$\mathbf{x}_t$
the same cell, unrolled: a deep network with tied weights
$\mathbf{s}_1$
$\mathbf{s}_2$
$\mathbf{s}_3$
$\mathbf{s}_4$
$\mathbf{h}_0$
$\mathbf{h}_1$
$\mathbf{h}_2$
$\mathbf{h}_3$
$\mathbf{h}_4$
f
f
f
f
$\cdots$
$\mathbf{x}_1$
$\mathbf{x}_2$
$\mathbf{x}_3$
$\mathbf{x}_4$